User Manual — Keychain Addon for Blender 5.0

Overview

This addon is designed for quickly creating 3D keychains from text in Blender 5.0. It builds raised text on the keychain, forms a base plate beneath the letters, adds an ear for a keyring, cuts a hole, and lets you export the finished model as an STL file.

Suitable for personalized keychains, test blanks, and base shape variations. The result is created as a mesh object directly in the Blender scene, where it can be refined manually or exported to STL right away.

Purpose

The addon generates a 3D keychain from text. It:

- Builds the top text on the keychain
- Forms a backing plate under the letters
- Adds an ear for a keyring
- Cuts a hole
- Exports the finished model as STL

Installation

1. Open Blender 5.0.
2. Go to **Edit** > **Preferences** > **Add-ons**.
3. Click **Install...** and select the addon `.py` file.
4. Enable the addon by checking the checkbox in the list.

After installation, the addon panel will appear in **3D Viewport** > **Sidebar** > **Keychain**. Interface sections can be collapsed and expanded like standard Blender subpanels.

Quick Start

1. Open the **Keychain** tab in the right sidebar of the 3D View.
2. In the **Text** section, enter the keychain text and select a font if needed.
3. In the **Base** section, set the width, letter spacing, and main base parameters. Letter spacing affects the distance between characters via `space_character`.
4. If needed, enable the ear and hole in the **Ear** and **Hole** sections.

5. Click **Build Joined Filled Pads** to generate the model.
6. When satisfied with the result, click **Save STL** to export.

Example: Enter a name, set the width to 50 mm, leave the default base parameters, and enable the left ear. This gives you a ready keychain blank that can be saved to STL immediately.

Interface

Text

The **Text** section controls the text content and font. This is where you set the inscription and the path to the TTF/OTF font file.

Parameters:

- **Text** — the keychain inscription.
- **Font Path** — path to the font file.
- **Choose Font** — button to select a font via file dialog.

Base

The **Base** section contains the most frequently used shape parameters. This is where you typically configure the overall size and character density.

Parameters:

- **Width (mm)** — the final width of the keychain in millimeters.
- **Letter Spacing** — character spacing; the higher the value, the farther apart the characters. Based on the `space_character` property.
- **Base Merge Distance** — vertex merge threshold when preparing the base.
- **Top Merge Distance** — vertex merge threshold for the top text.
- **Base Margin (mm)** — extra margin of the base plate around the letters.

Base Advanced

The **Base Advanced** section contains less common base settings. These are typically only changed for thin fonts, non-standard geometry, or a more expressive outline.

Parameters:

- **Base Thickness (mm)** — thickness of the bottom base plate.
- **Text Height (mm)** — height of the raised text.
- **Sharp Angle Limit (deg)** — threshold above which sharp corners may be softened.
- **Corner Inset (mm)** — amount of corner rounding/softening.
- **Corner Segments** — number of rounding segments.

Ear

The **Ear** section controls the side ear for the keyring. The ear can be placed on the left or right; both positions use the same set of parameters.

Parameters:

- **Add Side Ear** — enable or disable the ear.
- **Ear Side** — ear position: Left or Right.
- **Ear Height (mm)** — height of the ear.
- **Ear Length (mm)** — length of the ear.
- **Ear Y Offset (mm)** — vertical offset of the ear up or down.
- **Ear Gap (mm)** — gap between the ear and base before joining.
- **Ear Bevel (mm)** — rounding of the outer edge of the ear.
- **Ear Bevel Segments** — rounding detail level.
- **Intersection Search Step (mm)** — step size when joining the ear to the base.
- **Max Push Into Base (mm)** — maximum depth the ear extends into the base.

Hole

The **Hole** section controls the hole in the ear. For the right ear, the offset calculation is mirrored from the outer edge so that the behavior remains intuitive.

Parameters:

- **Create Ear Hole** — enable or disable the hole.
- **Hole Diameter (mm)** — diameter of the hole.
- **Hole Offset From Outer Edge (mm)** — distance from the outer edge of the ear to the nearest edge of the hole.
- **Hole Y Offset (mm)** — vertical offset of the hole.
- **Hole Segments** — number of segments in the cylinder used to cut the hole.

The hole is cut using a Boolean **DIFFERENCE** operation. In Blender, this uses a Boolean modifier that removes the volume of intersection between the main object and the cutter object.

Checks

The **Checks** section manages pre-extrusion checks on the base. These parameters are useful when you need to catch problematic overhanging faces.

Parameters:

- **Overhang Angle (deg)** — overhang angle for the check.
- **Delete Overhang Faces Before Extrude** — automatically delete detected problematic faces.
- **Hide Helpers** — hide intermediate helper objects after the build.

Export

The **Export** section contains the main model interaction buttons. It may also include a settings reset button, depending on your current addon version.

Buttons:

- **Build Joined Filled Pads** — build the keychain.
- **Save STL** — save the active mesh object to STL.
- **Reset All Settings** — reset all parameters to factory defaults (if present in your build).

Recommended Workflow

For most use cases, the following order works well:

1. Enter the text.
2. Select a font.
3. Set **Width (mm)**.
4. Adjust **Letter Spacing** if needed.
5. Build the model.
6. Enable the ear and hole if required.
7. Click **Build** again.
8. Save the result to STL.

If the font is thin or decorative, adjust **Base Margin** first before moving on to **Base Advanced**. This typically gives the most predictable result for the base plate.

Default Values

Exact values depend on your current version of the addon file, but the typical starting configuration is: width 50 mm, letter spacing 1.0, base thickness 3 mm, text height 2 mm, ear enabled, hole enabled. Letter spacing in Blender allows a range of 0–10 for `space_character`.

If you have already changed the defaults in the code, treat those as the factory values for your build. The **Reset All Settings** button is the most convenient way to restore them, if it has been added.

Practical Tips

- For short names, a width of 45–60 mm is usually sufficient.
- If letters are merging together, increase **Letter Spacing** and if needed **Base Margin**. The `space_character` property is specifically designed to control inter-character spacing.

- If the base plate looks too bulky, reduce **Base Margin**.
- If the text appears visually too recessed, increase **Text Height**.
- If the ear looks rough, increase **Ear Bevel Segments**.
- For the right ear, you do not need to manually recalculate the hole offset — the offset-from-outer-edge parameter already works in a mirrored fashion.

Known Behavior

Occasionally, rebuilding with the same parameters may produce a slightly different base shape. This is because the base generation and subsequent geometry processing can yield slightly different results in sensitive areas.

In practice, this can be treated as a feature of the generator — it sometimes produces interesting base variations. If you need to preserve a specific result, simply save it as a separate object or export it to STL immediately.

Exporting to STL

1. Make sure the build completed successfully.
2. Activate the finished keychain object if required.
3. Click **Save STL**.
4. Choose a folder and filename.
5. Confirm the save.

STL is suitable for 3D printing, sending to a slicer, or archiving the final model. The Boolean hole and joined geometry are already included in the exported mesh object.

Resetting Settings

If your version includes the **Reset All Settings** button, it restores all addon parameters to the factory values defined in the code. This is convenient after experimenting with fonts, the ear, and advanced base parameters.

Use the reset before starting a new model if you want to avoid any influence from previous settings. This is especially useful when working on a series of keychains with multiple names.

Typical Example

A basic scenario:

Parameter	Value
Text	Sergey

Parameter	Value
Width	50
Letter Spacing	1.0
Add Side Ear	Enabled
Ear Side	Left
Hole Diameter	3.0 mm
Hole Offset From Outer Edge	2.0 mm

This setup typically produces a standard personalized keychain with an ear and a keyring hole. From here, you only need to change the font, width, and ear side — leaving everything else as is.